

TROC16_423_ISA

Tagged Register Oriented Computer

p2 mnemonic prefix indicates a two byte, 16-bit instruction

?R column indicates if residue register updated

I suffix indicates immediate value of -15 to +15 or -32678 to +32678

little n to indicate the 5-bit N value in the instruction (-15 to +15)

TROC16 16-bit instructions, half word addressing only, N indicates 16-bit value in next

mnemonic	encoding	R?	operation
p2IN	000001	N	Store value of input port n into D
p2OUT	001001	N	Place D into output port n
p2LDT	000001	N	When n is "10000" load D from memory location N
p2STT	001001	N	When n is "10000" store D into memory location N
p2BSR	010001	N	Branch to PC + n or N , save address of next inst in D
p2MOV	011001	N	Move register S to D
p2BRZ	100001	N	Branch to PC + n or to N if D is zero
p2BRNZ	101001	N	Branch to PC + n or to N if D is non-zero
P2BRP	110001	N	Branch to PC + n or to N if D is positive (MSB is zero)
P2BRM	111001	N	Branch to PC + n or to N if D is negative (MSB is one)
p2SHFT	000011	Y	if n not "10000" shift or adjust exponent by signed n
p2INSRT	000011	N	if n is "10000" insert field from residue into D per N
p2FUNC	001011		if n not "10000" do a one operand function $fn(D) \Rightarrow D$
p2EXTRCT	001011	~Y	if n is "10000" extract field from D and place in residue
p2AOBR	010011	N	Increment D and branch to N if D not zero
p2SOBR	011011	N	Decrement D and branch to N if D not zero
p2LDIU	100011	N	load unsigned immediate, set tag bits to unsigned
p2LDIS	101011	N	load signed data immediate, set tag bits to signed
p2LDIF	110011	N	load float data immediate, convert N to register format
p2LDIF2	111011	N	load float2 data immediate, convert N to register format
p2ADD	000101	Y	Add S to D
p2SUB	001101	Y	Subtract D from S
p2MUL	010101	Y	Multiply D by S and store in D
P2DIV	011101	Y	Divide D by S and store in D
p2AND	100101	N	Logical AND between S and D to D
p2OR	101101	N	Logical OR between S and D to D
p2XOR	110101	N	Logical exclusive or between S and D to D
p2CMP	111101	~Y	S subtracted from D and placed in residue register
p2ADDI	000111	Y	Add I to D
p2SUBI	001111	Y	Subtract D from I
p2MULI	010111	Y	Multiply D by I and store in D
p2DIVI	011111	Y	Divide D by I and store in D
p2ANDI	100111	N	Logical AND between I and D to D
p2ORI	101111	N	Logical OR between I and D to D
p2XORI	110111	N	Logical exclusive or between I and D to D
p2CMPI	111111	~Y	D subtracted from I and placed in residue register

TROC16 8-bit instructions

mnemonic	encoding	R?	operation
p1TRAP	00000000	N	To catch a wayward instruction pointer
p1BKPT	01000000	N	Take a breakpoint trap
p1ALGN	10000000	N	Force program counter to next word address
p1ALGNC	11000000	N	Force program counter to next cache line address

TROC16 24-bit instructions

mnemonic	encoding	R?	operation
p3LDU16	00 000110	N	Load half-word unsigned
p3LD16	00 010110	N	Load half-word signed
p3FLD16	00 100110	N	Load half-word float
p3F2LD16	00 110110	N	Load half-word float2
p3LDU32	00 001110	N	Load half-word and its tag
p3LD32	00 011110	N	Load half-word and its tag
p3FLD32	00 101110	N	Load half-word and its tag
p3F2LD32	00 111110	N	Load half-word and its tag
p3ST16	01 000110	N	Store half-word
p3ST32	01 001110	N	Store half-word and its tag

TROC16 16-bit FUNC instructions

mnemonic	encoding	R?	operation
p2SETUS	00000	N	set D to unsigned
p2SETS	00001	N	set D to signed
p2SETFLT	00010	N	set D to float
p2SETFLT2	00011	N	set D to float2
p2ABS	00100		take the absolute value of D
p2NABS	00101		take the negative absolute value of D
p2fTAG	00110	Y	place D's four tag bits into residue
p2fITAG	00111	N	place residue four LSB bits into D's tag
p2fLDUS	01100		Load residue with unsigned from M(D)
p2fLDUS	01101		Load residue with signed from M(D)
p2fLDUS	01110		Load residue with float from M(D)
p2fLDUS	01111		Load residue with float2 from M(D)
p2fST	10001	~Y	convert and store Residue to M(D)
p2CVTUS	11100	~Y	convert D to unsigned
p2CVTS	11101	~Y	convert D to signed
p2CVTFLT	11110	~Y	convert D to float
p2CVTFLT2	11111	~Y	convert D to float2
			Room for 14 more functions

2677 or a float converted to register format

.5) and big N (little n is “10000”)

memory half-word

more detail
Negative n signifies internal ports, Positive n external ports.
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Tag bits loaded from memory location $N+1$
Tab bits stored into memory location $N+1$
TROC24 always uses relative addressing
tag bits moved as well
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N format: uu0000www00mmmm data type, bit width, starting bit
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n is 2's complement
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N is in memory format
N is in memory format
Floating-point ADD makes use of & may modify the exponent tag bits
Floating-point SUB makes use of & may modify the exponent tag bits
Floating-point MUL makes use of & may modify the exponent tag bits
Floating-point DIV makes use of & may modify the exponent tag bits
operation is irrespective of data types
operation is irrespective of data types
operation is irrespective of data types
Floating-point CMP makes use of & may modify the exponent tag bits
for floating op I converted to register format of D before add
for floating op I converted to register format of D before subtract
for floating op I converted to register format of D before multiply
for floating op I converted to register format of D before divide
operation is irrespective of data types
operation is irrespective of data types
operation is irrespective of data types
for floating op I converted to register format of D before subtract

